Nike Sales Analytics: Business Requirements Document

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Project type: Business analytics case study on a sales dataset (Jan to Dec 2024), delivered as a Power BI dashboard

1. Business Problem

Nike management wants to understand the sales performance by product and region so that they will know what are the underperforming segments and where they should focus more on. Today the sales data records are just a transactional level record. From this the manager won't be able to know which region and product drives the revenue, which earns healthy profit, and whether revenue is going quarter to quarter.

2. Objective

- Ensure management see a dashboard that shows revenue, profits, units, and growth for any quarter, etc.
- Compare actual revenue with the allocated budget for each region.
- Show each region and each product contribution to total revenue and each region contribution to profit.
- Identify the underperforming regions and products, and revenue share and profit share differ
- Validate the dashboard numbers with SQL, and turn the findings into evidence-based recommendations.

In scope: regional, product, category and customer-group (Men, Women, Kids) sales analysis, quarterly growth, profit margin, budget comparison, discount analysis, one Power BI dashboard and SQL validation.
Out of scope: forecasting, customer-level analysis, pricing changes, live data refresh.

3. Stakeholders

Role	What they care about?	Decision they make	KPI needed
Business Head	Overall growth and profitability	Where to invest or cut off	Total Revenue, Total Profit, Margin%, Revenue Growth %
Regional Sales Manager	How their region performs compared to others	Where to focus or put more efforts on which stock	Regional Contribution, Revenue and profit
Sales Manager	Volume and at what price sold	Demand and supply, Target price, discount	Avg Selling Price, Total Units
Marketing Manager	Which products are performing better	Which product to promote more often	Product Contribution%, Profit Margin%
Data/BI Team	The authenticity and accuracy of data	How the dashboard is build and validated	All KPIs; match against SQL

4. Current Situation

- Sales are recorded at order level with date, region, product, units, revenue and profit.
- The dataset covers six regions (Bangalore, Delhi, Hyderabad, Kolkata, Mumbai, Pune), 19 products and 5 product categories (for example Basketball, Lifestyle, Running, Soccer and Training).
- Total revenue for 2024 is about 483K, and Q4 alone contributes about 275K of it. Q4 alone brings 275K (57% of the year), so results depend heavily on the quarter selected.
- The dataset is small. Q1 has 2 orders, Q2 has 3 orders and most products have orders from 1-4. Shares and growth rate is thus indicative and not conclusive
- There is no single view that combines regional contribution, product contribution, profit and quarter-on-quarter growth. This gap is what the dashboard fills. *(Assumption: no such view existed before this project.)*

5. Functional Requirements

ID	Requirement	Priority
FR-01	The dashboard shall display Total Revenue, Total Units, Average Selling Price and Profit Margin % as headline cards.	Must
FR-02	The dashboard shall show each region's share of revenue as a ranked bar chart.	Must
FR-03	The dashboard shall show each product's share of revenue as a ranked bar chart.	Should
FR-04	The dashboard shall show each region's share of profit alongside its revenue share, so differences are visible.	Should
FR-05	The dashboard shall show quarter-on-quarter Revenue Growth %, displayed as blank where there is no earlier quarter (Q1).	Must
FR-06	Users shall be able to filter every visual by Quarter, Region and Product Category.	Must
FR-07	Contribution percentages shall add up to 100% across the items shown, and growth shall display in percentage format.	Must

ID	Requirement	Priority
FR-08	Dashboard values shall reconcile with independent SQL results for the same filters.	Should
FR-09	The SQL analysis shall compare actual revenue with allocated budget for each region, showing the difference and variance %.	Should
FR-10	The SQL analysis shall show how discount levels relate to revenue and profit, and which product categories rely most on discounts.	Should
FR-11	Data quality shall be checked before reporting: total orders, duplicate orders and missing values.	Should

The detailed user stories and Given/When/Then acceptance criteria for these requirements are in *User-Stories.md*. The SQL queries are in *SQL-Analysis.sql*.

6. KPIs

KPI	Formula	Why it matters
Total Revenue	Sum of revenue for the selected filters	Headline measure of sales performance
Total Units	Sum of units sold	Separates volume from price effects
Average Selling Price	Total Revenue ÷ Total Units	Shows whether revenue comes from volume or price
Profit Margin %	Average of the dataset's Profit Margin(%) column for the selected orders	Shows how profitable the selected orders are (Q4: 15.54%)
Revenue Growth % (QoQ)	(Revenue - Previous quarter revenue) ÷ Previous quarter revenue	Shows momentum; Q4 is 81.16% against Q3
Regional Contribution %	Region revenue ÷ Total revenue	Shows which markets drive sales (Q4: Pune 27.26%, Delhi 7.18%)
Product Contribution %	Product revenue ÷ Total revenue	Shows which products carry the business (Q4: Zoom Freak 15.98%)
Regional Profit Contribution %	Region profit ÷ Total profit	Compared with revenue share, shows where a region earns more or less profit than its sales suggest
Budget Variance %	(Revenue - Budget) ÷ Budget	Shows whether a region meets its allocated budget (2024: five of six regions below budget)

7. Assumptions and Constraints

- The 2024 dataset is complete and accurate. Checked 40 orders, one duplicate order(removed), no missing revenue, profit or region
- Revenue and profit are in single currency and no conversion needed
- The dataset is of calendar year starting from January to December.
- Profit Margin % is the average of the dataset's Profit Margin (%) column, used the same way in the dashboard and in SQL. This column is profit as a percentage of cost, so it reads higher than total profit ÷ total revenue (13.48% for 2024).
- Growth is only useful when a single quarter selected. With the full year selected it is meaningless.
- Contribution percentages depend on the filters applied, so a single region or product selected on its own shows 100%.
- Budget is treated as the allocated target for the order's region.
- This is a case-study dataset, so recommendations are illustrative and not a real Nike business decision.

8. Expected Business Outcome

- Managers can see in seconds which regions and products drive revenue and profit, instead of working it out from raw records.
- Underperformers are visible: Delhi and Mumbai each contribute only about 11% of 2024 revenue, and Mumbai has the lowest profit share (9.11%) against about 11.5% for Delhi.
- Budget gaps are clear: five of six regions finished below budget, with Mumbai (-3.5%) and Hyderabad (-3.2%) furthest below.
- Discounting can be reviewed: orders with no discount have the highest average revenue (about 14.3K, against 5.4K for low-discount and 11.1K for high-discount orders).
- Focus can be set on evidence: protect high-margin products such as Blazer Mid and SuperRep Go, and review low-margin ones such as Free RN and LeBron 20.
- Quarter-on-quarter tracking shows whether performance is improving, and that Q4 carries 57% of the year's revenue.

Next phase: add forecasting, a target-versus-actual view, and automated data refresh. Confirm the unit of the discount field and retest discounting against margin on a larger dataset.